
THEORETICAL NEUROSCIENCE

TD4: EXCITATORY-INHIBITORY NETWORKS

All TD materials will be made available at https://github.com/yfardella/Theo_Neuro_TD_26_27.

In this tutorial, we will reproduce two results from Murphy & Miller (Neuron, 2009), Balanced Amplification: A New Mechanism of Selective Amplification of Neural Activity Patterns. "In cerebral cortex, ongoing activity absent a stimulus can resemble stimulus-driven activity in size and structure. In particular, spontaneous activity in cat primary visual cortex (V1) has structure significantly correlated with evoked responses to oriented stimuli. This suggests that, from unstructured input, cortical circuits selectively amplify specific activity patterns. Current understanding of selective amplification involves elongation of a neural assembly's lifetime by mutual excitation among its neurons. We introduce a new mechanism for selective amplification without elongation of lifetime: "balanced amplification.""

1 Balanced Amplification

We consider an excitatory-inhibitory network presented in the Fig. 1 and that can be described by the following linear rate model:

$$\tau \frac{dr}{dt} = -r + \underline{W}r - \underline{I}.$$

We have

$$\underline{r} = \begin{bmatrix} r_E(t) \\ r_I(t) \end{bmatrix}$$

where r_E and r_I are the average firing rates of the E and I populations respectively,

$$\underline{I} = \begin{bmatrix} I_E \\ I_I \end{bmatrix}$$

where I_E and I_I are the external inputs into the E and I populations respectively and

$$\underline{W} = \begin{bmatrix} w & -k_I w \\ w & -k_I w \end{bmatrix}$$

where $\underline{W}$ is the connectivity matrix. In the following, we assume that inhibition balances excitation, hence, $k_I > 1$.

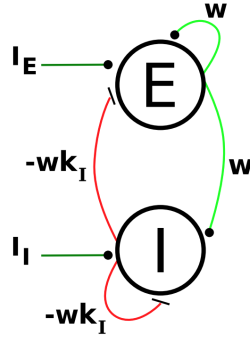


Figure 1

1.1 Network relaxation

In this subsection, we assume that there is no input into the populations, i.e. $\underline{I} = \underline{0}$ and that $r_E(0) = 1$ and $r_I(0) = 0$.

1. Find the eigenvalues and eigenvectors of $\underline{W}$. Comment on the sign of the eigenvalues.

These eigenvectors being non-orthogonal, the interpretability of the network dynamics in terms of the eigenvectors can be deceptive. Therefore, we propose to study the dynamics in the following orthogonal basis, letting,

$$\underline{p}^+ = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

and

$$\underline{p}^- = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

2. If we consider a single excitatory population recurrently exciting itself with strength w , what are the possible steady-state solutions depending on the value of w .
3. Compute $\underline{W} \cdot \underline{p}^+$ and $\underline{W} \cdot \underline{p}^-$. Comment.
4. Decompose $\underline{r}(t)$ as a sum of these 2 vectors, $\underline{r}(t) = \underline{p}^+ r_+(t) + \underline{p}^- r_-(t)$
5. Compute the time course of $r_+(t)$ and $r_-(t)$.
6. Setting $k_I = 1.1$, draw the time course of $r_E(t)$ for $w = 3/4$ and $w = 4$.
7. What does balanced amplification mean?

1.2 Response to a pulse input

In this subsection, we assume that $r_E(0) = 0$, $r_I(0) = 0$ and the input to the network is a pulse at time 0:

$$\underline{I}(t) = \begin{bmatrix} \tau \delta(t) \\ 0 \end{bmatrix}.$$

8. How are the ODEs governing the time course of $r_+(t)$ and $r_-(t)$ obtained Q4 modified?
9. Compute the time course of $r_+(t)$ and $r_-(t)$. Comment.